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Connect to AdventureWorksLT

Explorador de objetos

Conectar

localhost (17.0.1135.8 de SQL Server - sa)

Bases de datos

Bases de datos del sistema

Instantáneas de bases de datos

AdventureWorks2025

EcommerceDB

AdventureWorksLT2025

Seguridad

Objetos de servidor

Replicación

Alta disponibilidad de Always On

Administración

Agente SQL Server (Agent XPs deshabilitado)

Generador de eventos XEvent

SQLQuery1.sql...25 (sa (76))

```
1  -- Verify key tables in AdventureWorksLT
2  SELECT TOP (5) CustomerID, FirstName, LastName
3  FROM SalesLT.Customer;
4
5  SELECT TOP (5) SalesOrderID, OrderDate, CustomerID
6  FROM SalesLT.SalesOrderHeader;
7
8  SELECT TOP (5) ProductID, Name, ListPrice
9  FROM SalesLT.Product;
```

100 %

No se encontraron problemas.

Línea: 9, Carácter: 23 SPC CRI

Resultados

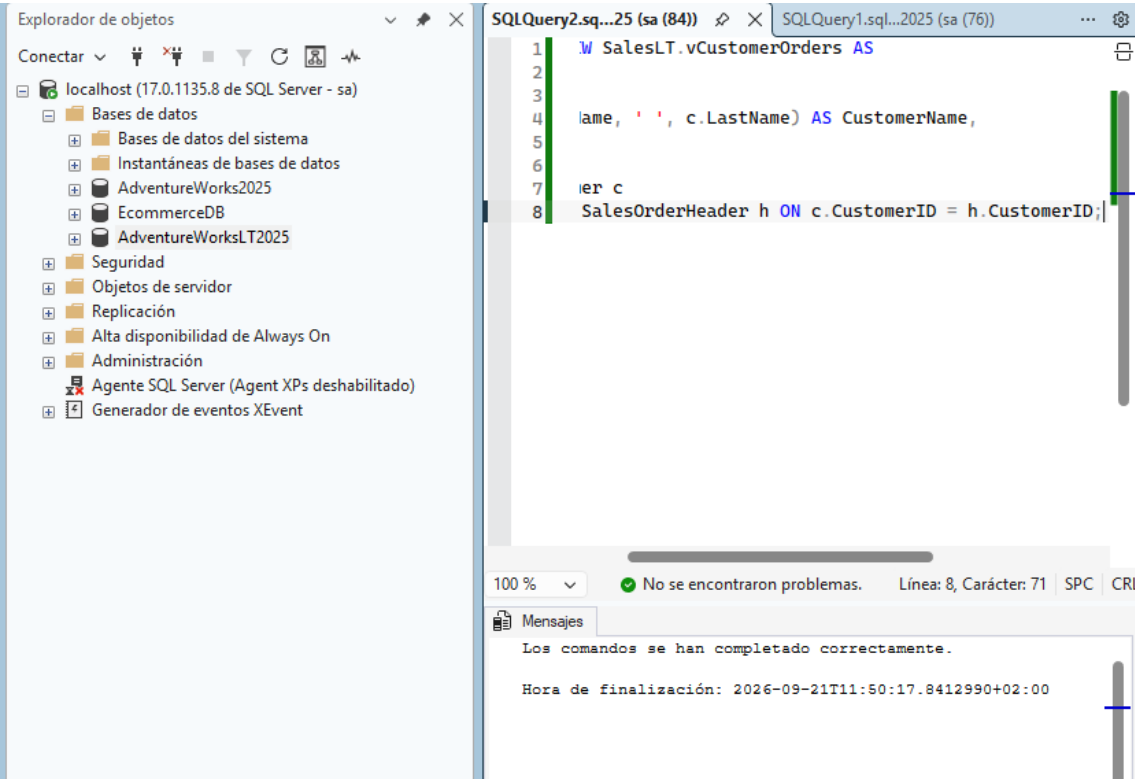
Mensajes

	CustomerID	FirstName	LastName
1	1	Orlando	Gee
2	2	Keith	Harris
3	3	Donna	Carreras
4	4	Janet	Gates
5	5	Lucy	Harrington

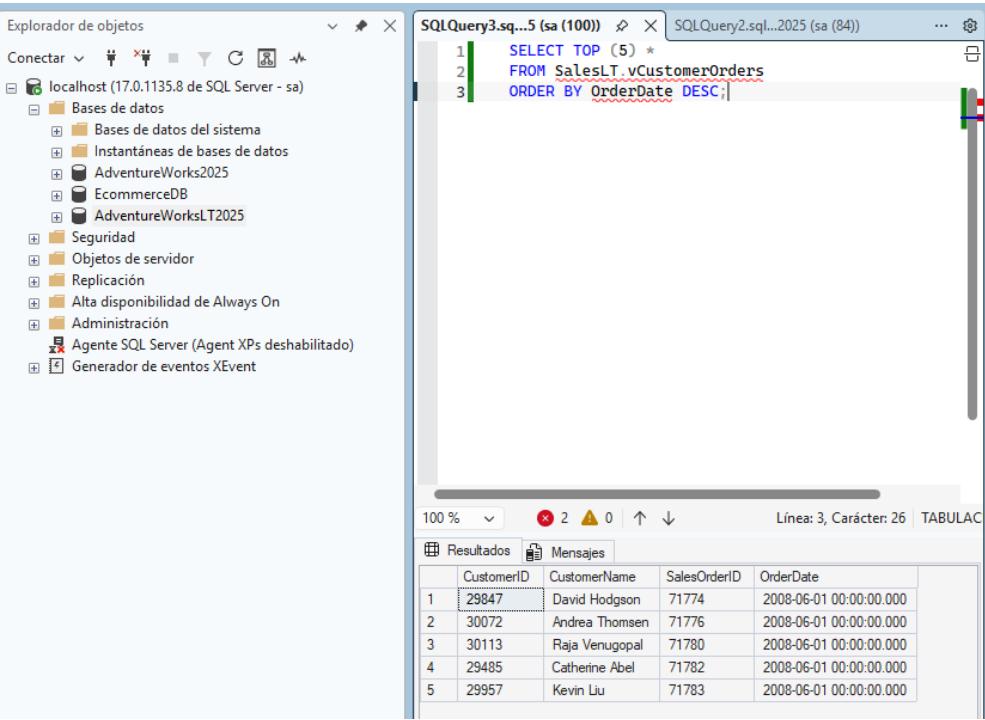
	SalesOrderID	OrderDate	CustomerID
1	71774	2008-06-01 00:00:00.000	29847
2	71776	2008-06-01 00:00:00.000	30072
3	71780	2008-06-01 00:00:00.000	30113
4	71782	2008-06-01 00:00:00.000	29485
5	71783	2008-06-01 00:00:00.000	29957

	ProductID	Name	ListPrice
1	680	HL Road Frame - Black, 58	1431,50
2	706	HL Road Frame - Red, 58	1431,50
3	707	Sport-100 Helmet, Red	34,99
4	708	Sport-100 Helmet, Black	34,99
5	709	Mountain Bike Socks, M	9,50

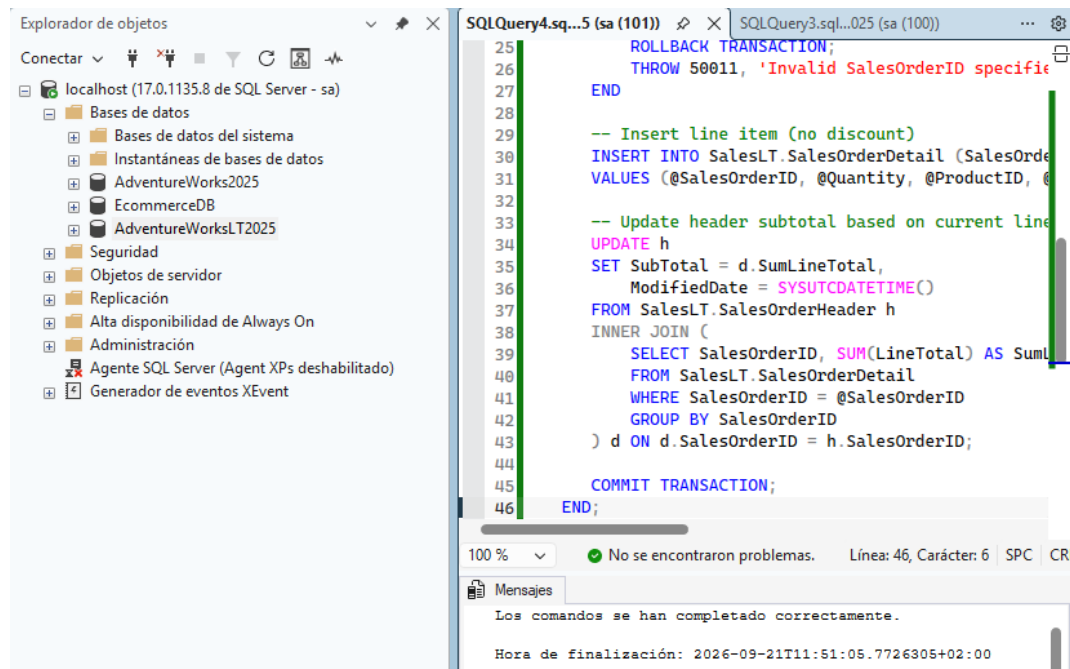
Create a view to simplify queries



Validate the view. Execute the following query.



Create a stored procedure to process an order.

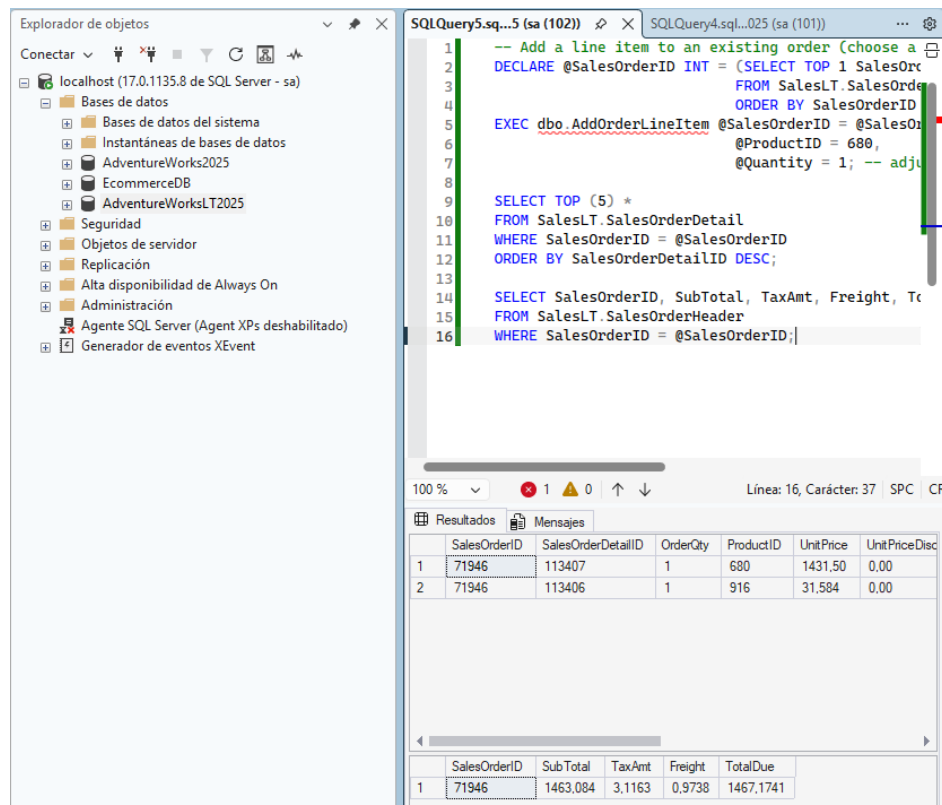


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Explorador de objetos' (Object Explorer) pane displays the server structure for 'localhost (17.0.1135.8 de SQL Server - sa)'. The 'Bases de datos' (Databases) folder is expanded, showing 'AdventureWorksLT2025'. The main pane shows a new SQL query window titled 'SQLQuery4.sql...5 (sa (101))'. The query contains the following T-SQL code:

```
25 ROLLBACK TRANSACTION;  
26 THROW 50011, 'Invalid SalesOrderID specified', 1;  
27 END  
28  
29 -- Insert line item (no discount)  
30 INSERT INTO SalesLT.SalesOrderDetail (SalesOrderID, ProductID, OrderQty, UnitPrice, UnitPriceDiscount)   
31 VALUES (@SalesOrderID, @ProductID, @Quantity, @UnitPrice, @UnitPriceDiscount);  
32  
33 -- Update header subtotal based on current line item  
34 UPDATE h  
35 SET SubTotal = d.SumLineTotal,  
36     ModifiedDate = SYSUTCDATETIME()  
37 FROM SalesLT.SalesOrderHeader h  
38 INNER JOIN (  
39     SELECT SalesOrderID, SUM(LineTotal) AS SumLineTotal  
40     FROM SalesLT.SalesOrderDetail  
41     WHERE SalesOrderID = @SalesOrderID  
42     GROUP BY SalesOrderID  
43 ) d ON d.SalesOrderID = h.SalesOrderID;  
44  
45 COMMIT TRANSACTION;  
46 END;
```

Below the code editor, the status bar indicates '100 %' zoom, 'No se encontraron problemas.' (No problems found), and 'Línea: 46, Carácter: 6' (Line: 46, Character: 6). The 'Mensajes' (Messages) pane at the bottom shows the message: 'Los comandos se han completado correctamente.' (The commands have completed successfully).

Execute the following T-SQL code to test the stored procedure.



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Explorador de objetos' (Object Explorer) pane displays the server structure for 'localhost (17.0.1135.8 de SQL Server - sa)'. The 'Bases de datos' (Databases) folder is expanded, showing 'AdventureWorksLT2025'. The main pane shows a new SQL query window titled 'SQLQuery5.sql...5 (sa (102))'. The query contains the following T-SQL code:

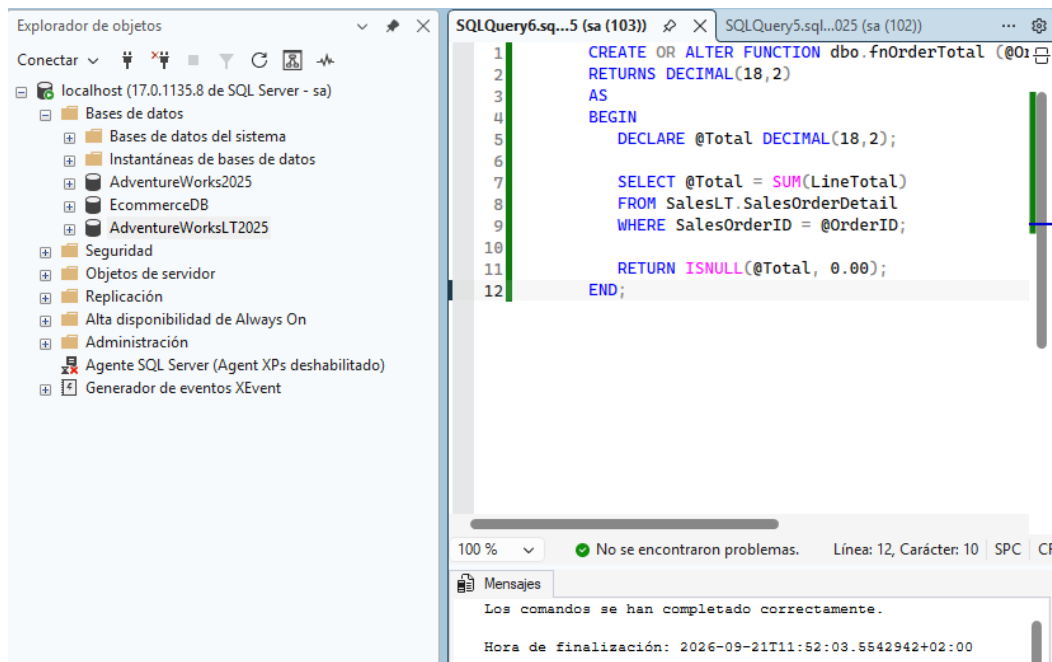
```
1 -- Add a line item to an existing order (choose a  
2 DECLARE @SalesOrderID INT = (SELECT TOP 1 SalesOrderID  
3 FROM SalesLT.SalesOrderHeader  
4 ORDER BY SalesOrderID  
5 EXEC dbo.AddOrderLineItem @SalesOrderID = @SalesOrderID,  
6 @ProductID = 680,  
7 @Quantity = 1; -- adjust quantity  
8  
9 SELECT TOP (5) *  
10 FROM SalesLT.SalesOrderDetail  
11 WHERE SalesOrderID = @SalesOrderID  
12 ORDER BY SalesOrderDetailID DESC;  
13  
14 SELECT SalesOrderID, SubTotal, TaxAmt, Freight, TotalDue  
15 FROM SalesLT.SalesOrderHeader  
16 WHERE SalesOrderID = @SalesOrderID;
```

Below the code editor, the status bar indicates '100 %' zoom, '1' error, '0' warnings, and 'Línea: 16, Carácter: 37' (Line: 16, Character: 37). The 'Resultados' (Results) pane at the bottom shows the results of the query:

SalesOrderID	SalesOrderDetailID	OrderQty	ProductID	UnitPrice	UnitPriceDiscount
71946	113407	1	680	1431.50	0.00
71946	113406	1	916	31.584	0.00

SalesOrderID	SubTotal	TaxAmt	Freight	TotalDue
71946	1463.084	3.1163	0.9738	1467.1741

Create a scalar function for reusable calculations



Execute the following T-SQL code to use the function:

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SQLQuery7.sql...25 (sa (53))

SQLQuery6.sql...025 (sa (103))

```
1 SELECT d.SalesOrderID, dbo.fnOrderTotal(d.SalesOrderID)
2 FROM SalesLT.SalesOrderDetail d
3 GROUP BY d.SalesOrderID
4 ORDER BY d.SalesOrderID DESC;
```

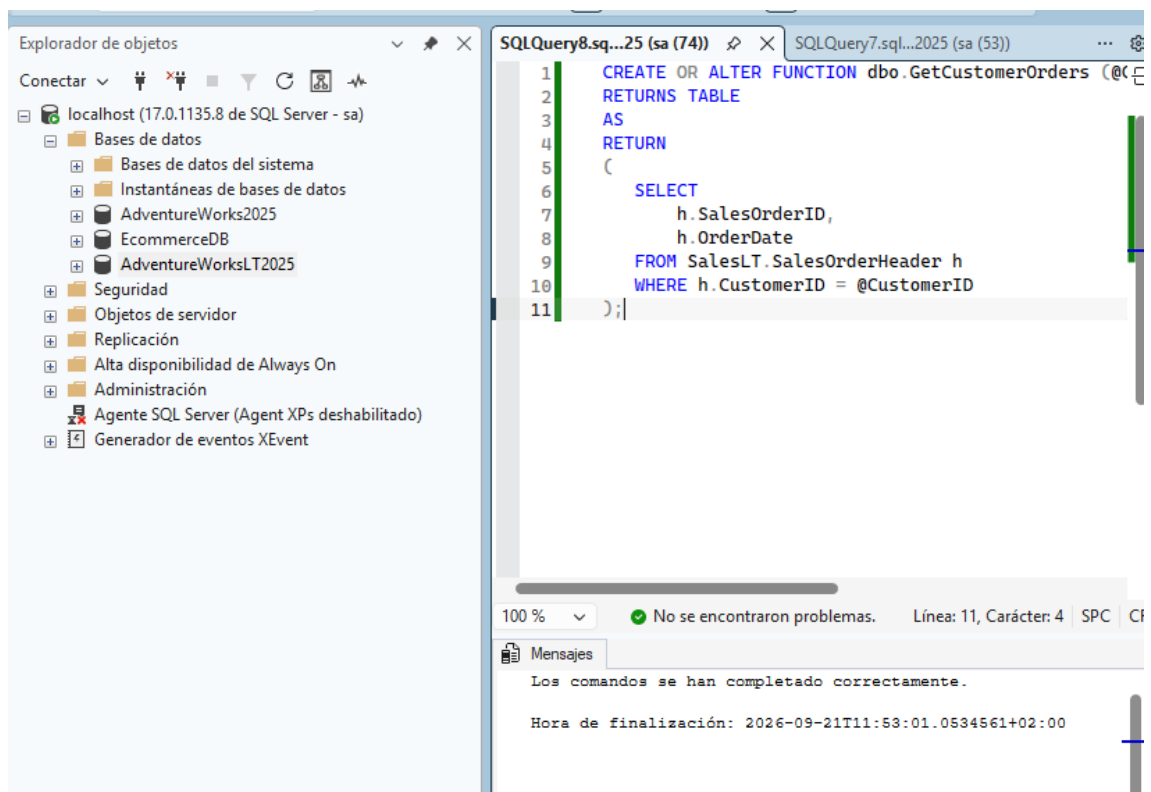
100 % 1 0 Línea: 4, Carácter: 31 TABULAC

Resultados Mensajes

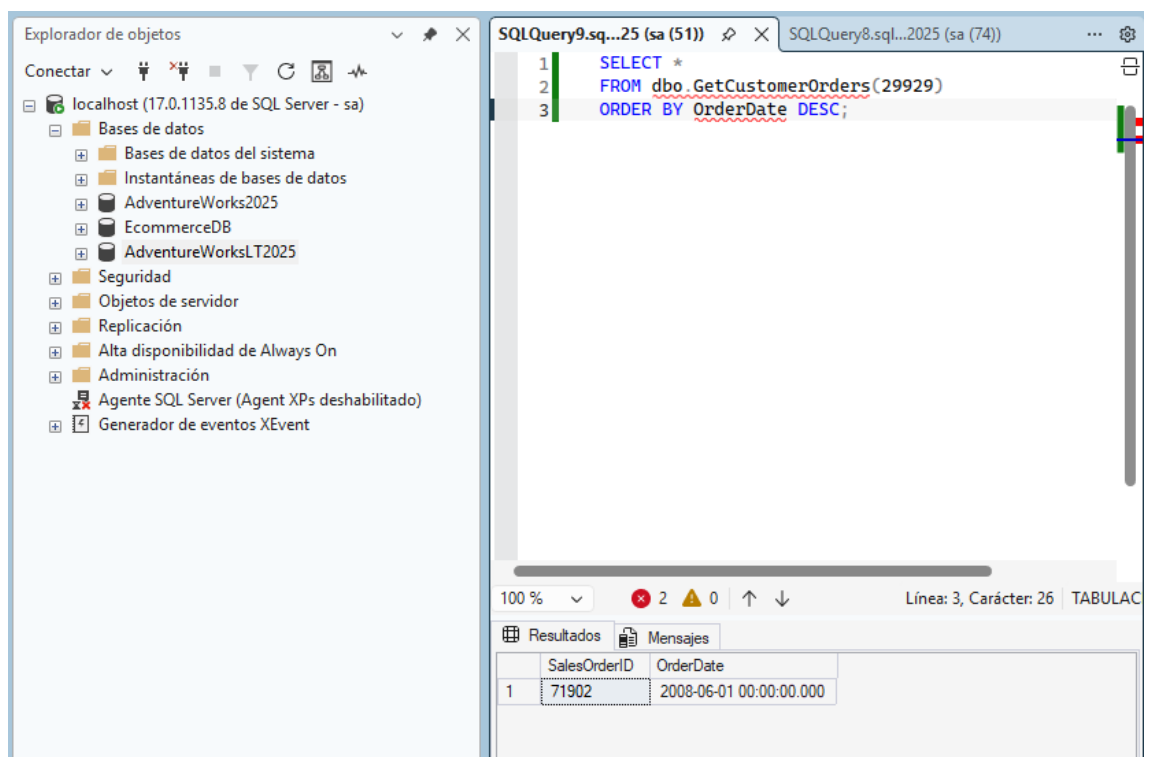
	SalesOrderID	OrderTotal
1	71946	1463.08
2	71938	74160.23
3	71936	79589.62
4	71935	5533.87
5	71923	96.11
6	71920	2527.13
7	71917	37.76
8	71915	1732.89
9	71902	59894.21
10	71899	1856.21
11	71898	53248.69
12	71897	10585.05
13	71895	221.26
14	71885	524.66
15	71867	858.90
16	71863	2777.14
17	71858	11528.84
18	71856	500.30

(17.0 RTM) sa (53) AdventureWorksLT2025 00:00:00 Fila: 1, Columna: 1 32 filas

Create an inline table-valued function (TVF)



Execute the following T-SQL code to query the function.



Execute the following T-SQL code to join the function to customers:

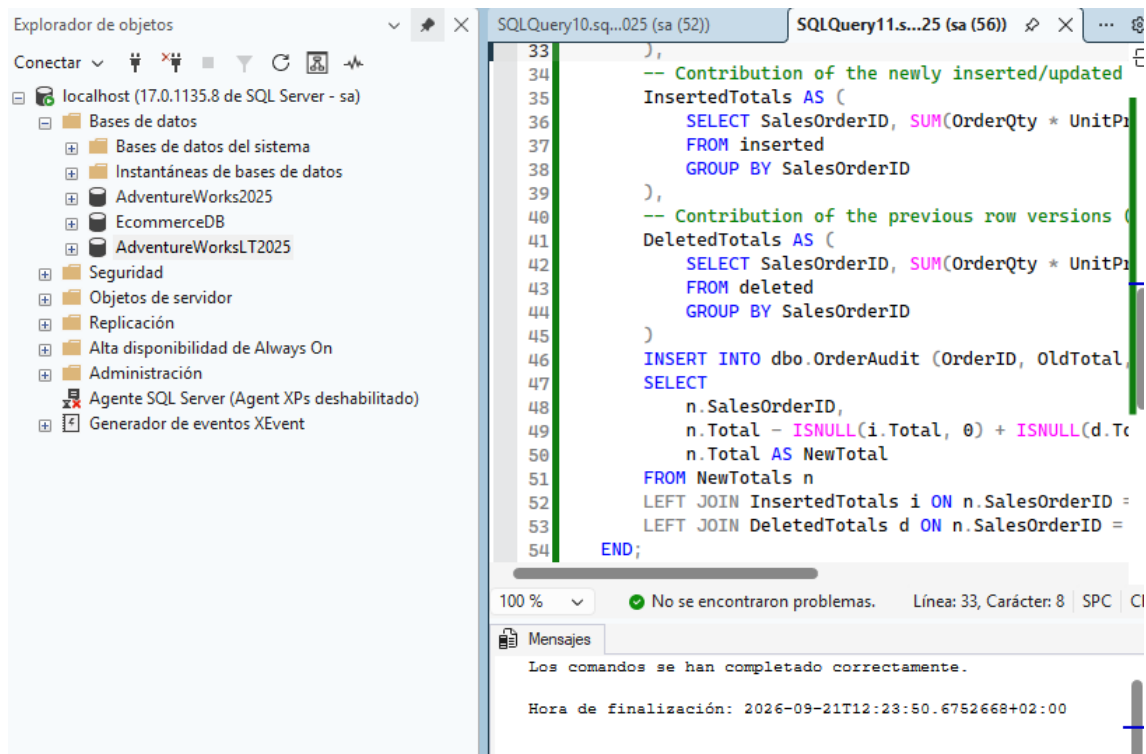
The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Explorador de objetos' (Object Explorer) pane displays the server structure for 'localhost (17.0.1135.8 de SQL Server - sa)'. The 'Bases de datos' (Databases) folder is expanded, showing 'AdventureWorksLT2025'. The main pane displays a T-SQL query in the 'SQLQuery10.s...25 (sa (52))' window. The query is as follows:

```
1 SELECT CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName
2 FROM SalesLT.Customer c
3 CROSS APPLY dbo.GetCustomerOrders(c.CustomerID) AS CustomerOrders
4 WHERE c.CustomerID = 29929;
```

Below the query editor, the 'Resultados' (Results) pane shows the output of the query. It displays a table with the following data:

	CustomerName	SalesOrderID	OrderDate
1	Jeffrey Kurtz	71902	2008-06-01 00:00:00.000

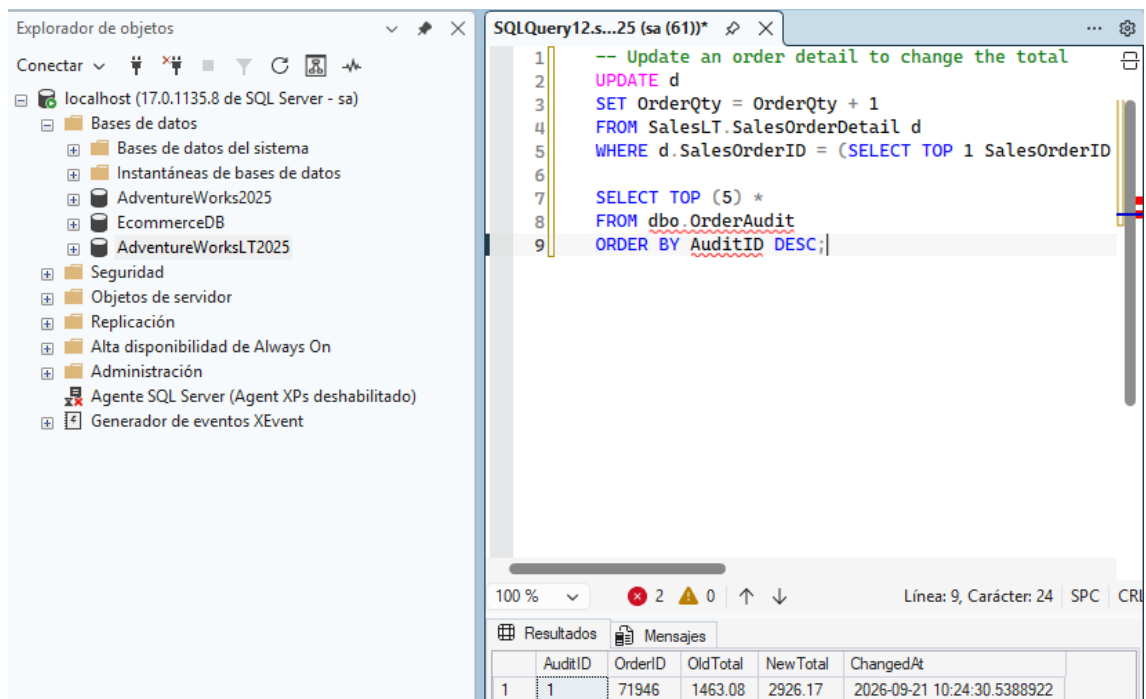
Create a trigger to log changes



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Explorador de objetos' (Object Explorer) pane displays the server hierarchy for 'localhost (17.0.1135.8 de SQL Server - sa)', including 'Bases de datos' (Databases) and 'Seguridad' (Security). The main window shows a SQL query script for 'SQLQuery11.s...25 (sa (56))'. The script is a trigger that logs changes to the 'OrderAudit' table. It includes comments for 'Contribution of the newly inserted/updated' and 'Contribution of the previous row versions'. The script uses 'INSERT INTO' and 'SELECT' statements to log changes. The status bar at the bottom indicates 'No se encontraron problemas.' (No problems found) and 'Línea: 33, Carácter: 8' (Line: 33, Character: 8). The 'Mensajes' (Messages) pane shows the execution results: 'Los comandos se han completado correctamente.' (The commands have completed successfully) and 'Hora de finalización: 2026-09-21T12:23:50.6752668+02:00' (Completion time: 2026-09-21T12:23:50.6752668+02:00).

```
33 -- Contribution of the newly inserted/updated
34 -- Contribution of the newly inserted/updated
35 InsertedTotals AS (
36     SELECT SalesOrderID, SUM(OrderQty * UnitPrice) AS NewTotal
37     FROM inserted
38     GROUP BY SalesOrderID
39 ),
40 -- Contribution of the previous row versions (
41 DeletedTotals AS (
42     SELECT SalesOrderID, SUM(OrderQty * UnitPrice) AS OldTotal
43     FROM deleted
44     GROUP BY SalesOrderID
45 )
46 INSERT INTO dbo.OrderAudit (OrderID, OldTotal, NewTotal, ChangedAt)
47 SELECT
48     n.SalesOrderID,
49     n.Total - ISNULL(i.Total, 0) + ISNULL(d.Total, 0) AS NewTotal
50     FROM NewTotals n
51     LEFT JOIN InsertedTotals i ON n.SalesOrderID = i.SalesOrderID
52     LEFT JOIN DeletedTotals d ON n.SalesOrderID = d.SalesOrderID
53 END;
```

Execute the following T-SQL code to test the trigger:



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Explorador de objetos' (Object Explorer) pane displays the server hierarchy for 'localhost (17.0.1135.8 de SQL Server - sa)'. The main window shows a SQL query script for 'SQLQuery12.s...25 (sa (61))*'. The script is a test query that updates an order detail to change the total. It includes an 'UPDATE' statement and a 'SELECT' statement to retrieve the updated data. The status bar at the bottom indicates 'Línea: 9, Carácter: 24' (Line: 9, Character: 24). The 'Resultados' (Results) pane shows the execution results in a table with columns: 'AuditID', 'OrderID', 'OldTotal', 'NewTotal', and 'ChangedAt'. The table contains one row with values: '1', '71946', '1463.08', '2926.17', and '2026-09-21 10:24:30.5388922'.

```
1 -- Update an order detail to change the total
2 UPDATE d
3 SET OrderQty = OrderQty + 1
4 FROM SalesLT.SalesOrderDetail d
5 WHERE d.SalesOrderID = (SELECT TOP 1 SalesOrderID
6 FROM SalesLT.SalesOrder
7 WHERE SalesOrderID < 1000000000)
8
9 SELECT TOP (5) *
10 FROM dbo.OrderAudit
11 ORDER BY AuditID DESC;
```

AuditID	OrderID	OldTotal	NewTotal	ChangedAt
1	71946	1463.08	2926.17	2026-09-21 10:24:30.5388922

